

Paper A-01: Earth Unified Environmental Transport: A Constraint-Driven Flux Dynamics (CDFD) Perspective

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Shared Part IV notation and evidence boundaries are defined in `PART_IV_METHODS_AND_CLAIM_BOUNDARY.md`. This paper states only its local observables, comparison, and failure conditions.

Abstract

This opening Earth-systems paper states the Part A modelling frame. It treats environmental transport as a CDFD/AFL hypothesis in which heat, water, nutrients, sediment, pollutants, and biological energy are represented as domain-specific fluxes Φ moving against constraints C through surfaces with responsiveness S and structural memory M_s . The release-local runtime outputs are internal diagnostics and tests that could break the mapping, not field evidence of climate, hydrological, ecological, or geological claims.

1 Series Context

Part I sets the flow–constraint language used here [3]. Part II develops the Life Number and the origins-of-life use of the same notation [4]. Part III applies AFL to biology and medicine [5]. Part IV [6], DOI <https://doi.org/10.5281/zenodo.20395901>, uses this Earth-systems opening paper to set the measurement boundary for the rest of Part A.

2 Earth-System Transport Frame

Earth systems are coupled transport networks. Atmospheric circulation moves heat and moisture; oceans redistribute heat and salinity; rivers move water and sediment; soils regulate nutrients and organic matter; ecosystems route biomass and information through trophic structure. CDFD/AFL does not replace the standard equations in these fields. It supplies a common audit layer for asking where flow rises, where capacity tightens, where

responsiveness is lost, and where memory keeps a system from returning to its earlier state.

3 Candidate Variable Map

CDFD term	Earth-system reading	Example proxy
Φ	environmental drive or throughput	heat flux, runoff, nutrient load
C	active constraint or capacity burden	aridity, channel resistance, toxicity
S	responsiveness or buffering	biodiversity, soil structure, wetland storage
M_s	structural memory	degradation, hysteresis, legacy contamination
Ψ_s	operating ratio	balanced, constrained, or overloaded regime

This map is hypothesis-level until the proxies, scales, datasets, and failure conditions are specified for a given Earth-system question.

4 Part A Sequence

The Part A papers move from broad environmental transport into atmosphere, ocean, hydrology, soil, ecology, biodiversity, pollution, agriculture, sustainability, collapse, synthesis, plate tectonics, and volcanology. The sequence is ordered to start with transport structure, then add specific environmental constraints and failure modes.

5 Measurement Layer

6 Current Runtime Diagnostic

The release-local discovery script includes selected Earth-system adapter rows for climate and atmosphere, ocean transport, hydrological flow, biodiversity, and pollution. The accompanying network-cascade diagnostic checks finite-value behavior in an overload and memory-locking stress test. I use those rows as a runtime audit, not as a replacement for Earth-system data.

7 Falsification Targets

The Earth-systems framing weakens if a domain cannot define measurable proxies for Φ , C , S , and M_s , if the proposed operating ratio fails to predict overload or recovery behavior, or if a standard domain model explains the same data without any added value from the CDFD/AFL mapping.

8 Cross-Domain Model Upgrade: Mechanism, Scale, and Tests

8.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For Earth Unified Environmental Transport, the proposed drive is radiative, hydrological, biogeochemical, and human forcing. The active limitation is transport resistance and finite environmental capacity. Responsiveness is carried by circulation, storage, ecological buffering, and rerouting, while retained state is represented by soil, ice, ocean, ecological, and contamination legacies. The outcome to explain is redistribution, overload, recovery, and regime change.

Table 1: Operational CDFD/AFL map for Paper A-01.

Term	Paper-specific observable meaning	Measurement question
Φ	radiative, hydrological, biogeochemical, and human forcing	What rate, load, gradient, or demand enters the focal system?
C	transport resistance and finite environmental capacity	Which measured bottleneck limits transfer, function, or recovery?
S	circulation, storage, ecological buffering, and rerouting	Which process changes effective capacity or reroutes the load?
M_s	soil, ice, ocean, ecological, and contamination legacies	Which retained state makes equal present loads produce different futures?
Y	redistribution, overload, recovery, and regime change	Which outcome is predicted out of sample, with uncertainty?

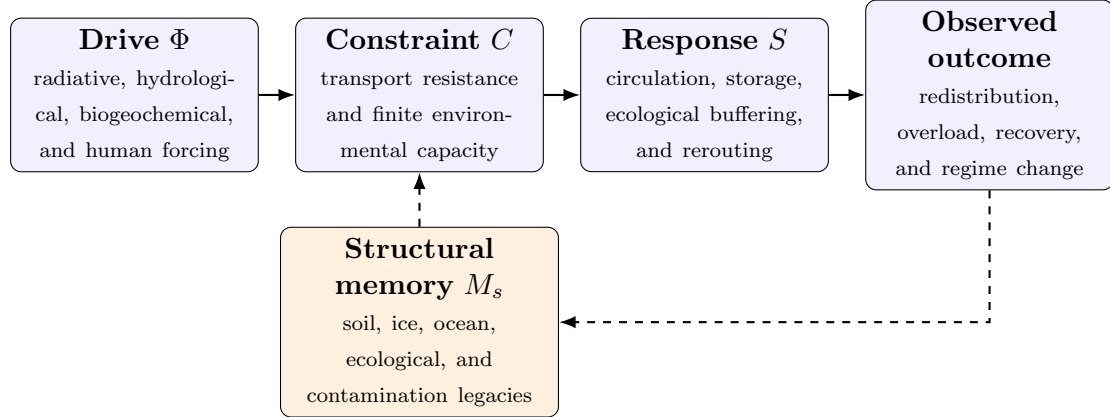


Figure 1: Paper A-01 observable CDFD/AFL architecture for Earth Unified Environmental Transport. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

8.2 Paper-Specific Causal Hypothesis

For this paper, the hypothesized causal sequence is specific. A change in radiative, hydrological, biogeochemical, and human forcing arrives faster than circulation, storage, ecological buffering, and rerouting can compensate. This raises or redistributes transport

resistance and finite environmental capacity. If the load persists, the retained state encoded by soil, ice, ocean, ecological, and contamination legacies changes the next response, so a later event of equal magnitude can produce a different trajectory in redistribution, overload, recovery, and regime change. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

8.3 Cross-Scale Translation Without Mechanistic Erasure

For the Earth systems family, the cross-domain translation compares coupled transport, finite environmental capacity, buffering, and retained landscape or ecosystem state. It cannot erase conservation laws, geometry, or scale; it must expose where those domain equations create comparable overload and recovery structures. [2, 7, 1] The required scale declaration for this paper is hours to millennia; catchments to the coupled Earth system. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

8.4 Evidence Ladder and Discriminating Tests

Table 2: Evidence ladder for Paper A-01. Each rung can fail independently.

Test	Design	Discriminating result
Proxy audit	Measure coupled reanalysis, remote sensing, gauge records, and ecological time series and preregister how each record maps to Φ , C , S , M_s , and Y .	The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.
Load ramp	Apply or observe a compound heat, rainfall, and pollutant pulse while estimating the dominant constraint and response time.	A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint.
Recovery and memory	Match present load across units with different histories, then compare recovery paths.	The memory claim requires history-dependent divergence after current conditions are controlled.
Model comparison	Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.	The paper’s stated falsifier is: the added variables do not improve prediction beyond ordinary coupled transport budgets.

8.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from coupled reanalysis, remote sensing, gauge records, and ecological time series and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state

extension and test whether it improves prediction of redistribution, overload, recovery, and regime change. Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the cross-domain translation audit.

9 Conclusion

Part A uses CDFD/AFL as a shared transport language for Earth systems. The framework is useful only when it sharpens measurable hypotheses and failure conditions. It remains separate from operational climate, hydrological, ecological, agricultural, or geological advice.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Part-specific runtime diagnostics are under each Part's `outputs/` directory.

Release-wide code: `Part_E_Synthesis/supplementary/`.

Generated summaries: `Part_E_Synthesis/outputs/`.

Figures: `Part_E_Synthesis/figures/`.

10 Reference Layer

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